

Candidate Number \_\_\_\_\_

Fraction \_\_\_\_\_

**2018 International Biology Olympiad National Team Selection Camp Practical Test Questions****Third Examination Room**

All equipment and materials required for this examination have been placed on the table. Please check the items listed on the checklist. If anything is missing, please raise your hand immediately to report it to the examiner.

| Equipment 1 (Please leave it on the table during scene) |                                                      | Equipment 2 (One set per person, no reuse) |            |
|---------------------------------------------------------|------------------------------------------------------|--------------------------------------------|------------|
| name                                                    | changes) Quantity                                    | name                                       | Quantity   |
| Dissecting                                              | 1 plastic glove, 2                                   |                                            | 1 pair, 1  |
| microscope,                                             | face masks, 1                                        |                                            | piece, 5   |
| pointed                                                 | insect needle (inserted into a                       |                                            | sticks, 1  |
| forceps, dissecting disc, paper                         | dissecting tray), 1                                  |                                            | branch,    |
| Oily                                                    | pack of plastic                                      |                                            | 3 branches |
| rag, ballpoint                                          | droppers, 1 syringe, 1 cherry red cockroach (in a 50 |                                            | 2 pieces,  |
| pen, turtle                                             | mL centrifuge tube), 1 ice cube                      |                                            | 1 cup,     |
| Donut Turtle                                            | (in a plastic cup),                                  |                                            | 1 bottle   |
| timer                                                   | "A"), 1 reagent A (in an eye drop bottle, labeled    |                                            |            |
|                                                         | 1 reagent                                            |                                            | 1 bottle   |
|                                                         | B (in an eye drop bottle, labeled "B"),              |                                            |            |
|                                                         | physiological                                        |                                            | 1 bottle   |
|                                                         | saline (in a 15 mL centrifuge tube, labeled "S").    |                                            |            |

Important Notes

1. Materials, medicines, and equipment on the table will not be replenished after use. 2. This exam

paper (including cover and question paper) consists of 4 pages and must be returned in its entirety upon

submission. 3. Please ensure both cockroaches are alive (able to move); they will not be replaced during the competition.

4. Please be careful not to let the turtle be crushed by the equipment or

escape. 5. Please ensure the turtle is not under stress; it will not be replaced during the competition. 6.

Exercise caution when using the injection needle; after use, the protective cap must be properly closed; violations will result in point deductions.

## Question 1: Observation of cockroach sex and heartbeat (50%)

A. Place the cockroach under a dissecting microscope and observe the center of the dorsal side of its thorax or abdomen. A structure resembling a black line can be seen. This structure contracts and expands, and it is the heart (dorsal blood vessel). The cockroach's dorsal plate is thin and translucent, so the beating of its heart can be observed directly.

B. Commonly used anesthesia methods for cockroaches include carbon dioxide and hypothermia. This time, hypothermia will be used. Please place the cockroach in the provided ice cubes. After about five to ten minutes, the cockroach will become unconscious. Then, using pointed tweezers, place the cockroach face down on a dissecting tray and secure its wings with an insect pin to prevent movement. You can then begin observing the cockroach's heart. Please note:

(A) Please place the dissecting tray directly under the dissecting microscope to avoid sticking.

(B) If the operation causes the cockroach's heartbeat to stop, no replacement will be provided.

C. Please use the equipment and cockroaches provided on the table to carefully observe and answer the related questions.

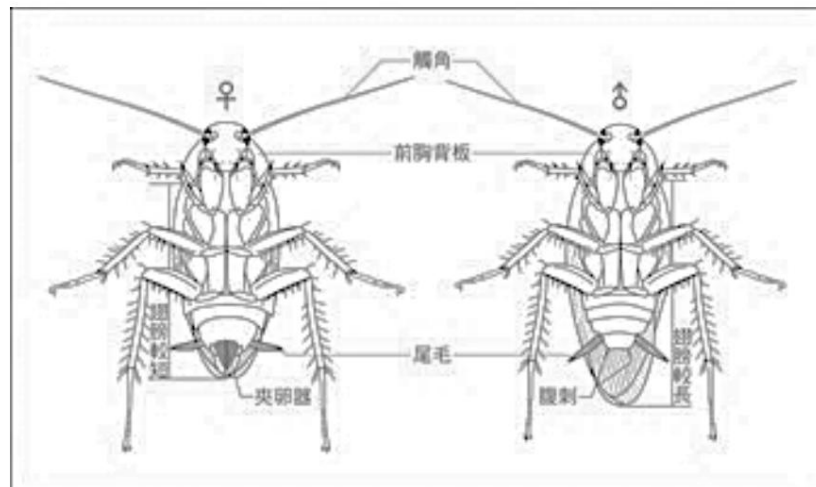


Figure 1. Appearance of a cockroach

1. Please determine the sex of the assigned cockroach. Mark V in the  $\checkmark$  and write down your reasoning. (10 points) (Please have the invigilator confirm and

grade your answer along with sub-question 2. )  $\checkmark$  Female  $\checkmark$  Male Reasoning:

Invigilator's Signature: 2. Please \_\_\_\_\_

operate under a dissecting microscope and locate the cockroach's heart, then move it to the center of your field of view. (5 points) (Please raise your

hand after completing the question. This answer, along with sub-question 1, will be confirmed and graded by the invigilator.)

Invigilator's Signature: \_\_\_\_\_

3. Please measure and present the average heart rate (beats/minute) of a cockroach in a table. (10 points)

4. Acetylcholine is a common neurotransmitter in the central and peripheral nervous systems, and it also participates in the regulation of heart rate. It is known that reagents A and B on the table contain acetylcholine prepared with physiological saline.

For alkaline solutions, please design an experiment and answer the following questions based on the results. Refer to Figures 2 and 3 for the administration

methods. 1. Compare the effects of injecting different solutions on the heart rate of

cockroaches. 2. The bottle number containing acetylcholine

solution. (Answers must include experimental design, experimental data, and experimental results; otherwise, this sub-question will not be graded.) (25 points)

Question 2: The Mechanism of a Tortoise Turning Over (50 points)

1. The spotted turtle and the donut turtle belong to the families Geoemydidae and Emydidae, respectively. The former is found in Southeast Asia (including Taiwan), while the latter is found in the southern United States.
2. Both are freshwater turtles of the cryptocerataceae family.
3. After observing these two turtles, design an experiment to explain the correlation between the turtle's morphology and its ability to easily and quickly right itself after being tipped over.
4. Please describe the phenomenon you want to test, the data you want to collect, and how you will obtain this data. And why this data can answer your question.